DUAL PROBLEMS

Thirty-sixth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Primal

$$\text{Maximize } z = 2x_1 + 3x_2,$$

subject to

$$x_1 + x_2 \le 8,$$
$$2x_1 + x_2 \le 6,$$
$$x_1, x_2 \ge 0$$

DUAL

$$\text{Minimize } z = 8y_1 + 6y_2,$$

subject to

$$y_1 + 2y_2 \ge 2,$$
$$y_1 + y_2 \ge 3,$$
$$y_1, y_2 \ge 0$$

Primal-Dual Relations

Primal

$$\text{Minimize } \mathbf{c}^T \mathbf{x}$$
$$\text{Subject to } \mathbf{Ax} \geq \mathbf{b}$$
$$\mathbf{x} \geq 0$$

Dual

$$\text{Maximize } \mathbf{y}^T \mathbf{b}$$
$$\text{S. t. } \quad \mathbf{y}^T \mathbf{A} \leq \mathbf{c}^T$$
$$\mathbf{y} \geq 0$$

Two equivalent forms

Primal

Minimize $\mathbf{c}^T \mathbf{x}$

Subject to $\mathbf{Ax}=\mathbf{b}$

$\mathbf{x} \geq 0$

Equivalent Primal

$$\text{Minimize } c^T x$$

$$\text{subject to } Ax \geq b$$

$$-Ax \geq -b$$

$$x \geq 0$$

A way of representation

- Suppose $\begin{bmatrix} A \\ -A \end{bmatrix}$ is reprsented by $[A]$.

$\begin{bmatrix} b \\ -b \end{bmatrix}$ is reprsented by $[b]$

Primal-Dual Relations

Primal

Minimize $\mathbf{c}^T \mathbf{x}$

Subject to

$$\begin{bmatrix} \mathbf{A} \\ -\mathbf{A} \end{bmatrix} \mathbf{x} \geq \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \end{bmatrix}$$

$\mathbf{x} \geq 0$

Dual

Maximize $(u,v)^T \begin{bmatrix} b \\ -b \end{bmatrix}$

S. t. $(u,v)^T \begin{bmatrix} A \\ -A \end{bmatrix} \leq \mathbf{c}^T$

$u,v \geq 0$

DUAL

Maximize $\mathbf{u}^T\mathbf{b} - \mathbf{v}^T\mathbf{b}$

S.t. $\mathbf{u}^T\mathbf{A} - \mathbf{v}^T\mathbf{A} \leq \mathbf{c}^T$

$\mathbf{u}, \mathbf{v} \geq \mathbf{0}.$

Let $\mathbf{y} = \mathbf{u} - \mathbf{v}$

Maximize $\mathbf{y}^T\mathbf{b}$

S.t. $\mathbf{y}^T\mathbf{A} \leq \mathbf{c}^T,$

y unrestricted

Making Dual from standard primal

Standard primal

Minimize $\mathbf{c}^T \mathbf{x}$

Subject to $\mathbf{Ax}=\mathbf{b}$

$\mathbf{x} \geq 0$

- Usually **b** is non-negative.

Dual

$$\text{Maximize} \quad \mathbf{y}^T\mathbf{b}$$

$$\mathbf{S.t.} \quad \mathbf{y}^T\mathbf{A} \leq \mathbf{c}^T,$$

$$\boldsymbol{y} \quad \text{unrestricted}$$

Important rules

- If the standard primal is of maximization type, the dual will be minimization type with $\geq$ type constraints and unrestricted variables.
- If the standard primal is of minimization type, the dual will be maximization type with $\leq$ type constraints and unrestricted variables.

Obtain the dual form of the following problem:

$$\text{Minimize } z = 40y_1 + 24y_2 + 10y_3$$

s.t.

$$y_1 + 3y_2 + y_3 \geq 2,$$
$$3y_1 + y_2 + y_3 \geq 5,$$
$$y_1 \geq 0, y_2 \geq 0, y_3 \geq 0.$$

Dual form:

$$\text{Maximize } w = 2x_1 + 5x_2$$

s.t

$$x_1 + 3x_2 \leq 40,$$
$$3x_1 + x_2 \leq 24,$$
$$x_1 + x_2 \leq 10$$
$$x_1 \geq 0, x_2 \geq 0.$$

Write down the dual form of the following problem:

Maximize $2x_1 + 5x_2$

s.t.

$$x_1 + 3x_2 \leq 40,$$
$$3x_1 + x_2 \leq 24,$$
$$x_1 + x_2 \leq 10,$$
$$x_1 \geq 0,\ x_2 \geq 0.$$

Dual form:

Minimize $W=40\ y_1+24\ y_2+10\ y_3$

subject to: $y_1+3y_2+y_3\geq 2$

$3y_1+y_2+y_3\geq 5$

$y_1,y_2,y_3\geq 0$

Primal:

Minimize $z = x_1 + 3x_2$

Subject to

$x_1 + x_2 = 4,$

$x_1 \geq 0, x_2 \geq 0.$

Dual:

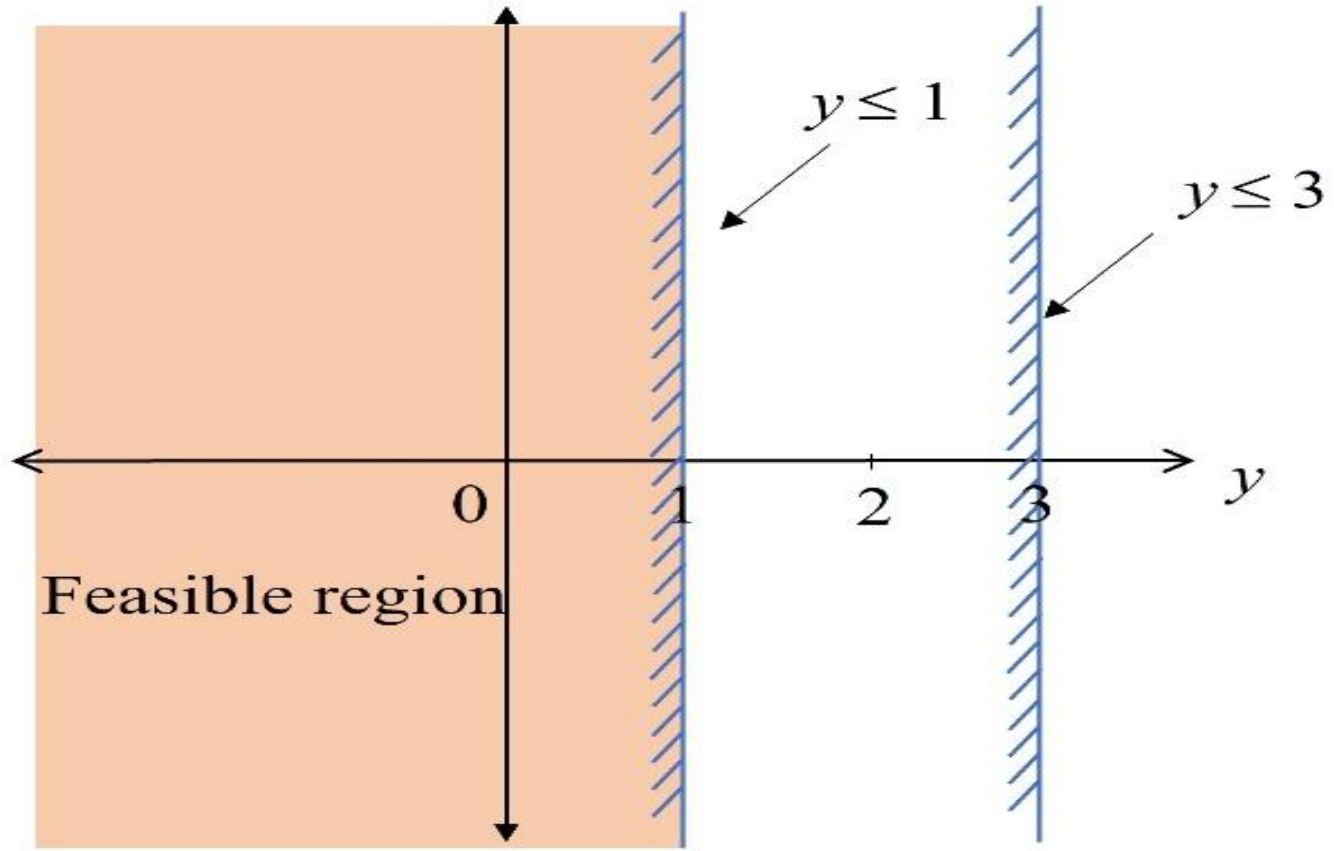
Maximize $w = 4y$

Subject to

$y \leq 1,$

$y \leq 3,$

y is unrestricted.


$y \leq 1$
$y \leq 3$
0
1
2
3
y
Feasible region

Maximize $z = x_1 + 2x_2 + 3x_3$

Subject to

$$\frac{x_1}{x_2 + x_3} \geq 0.5,$$

$$3x_1 + 2x_2 + x_3 \leq 6$$

$$x_1, x_2, x_3 \geq 0.$$

Maximize $z = 50x_1 + 40x_2 + 30x_3 + 25x_4 + 20x_5 + 10x_6$

subject to

$$5x_1 + 2x_2 + x_3 + x_4 + x_5 + 2x_6 \le 10,$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \ge 0.$$